Design System

PopNotes

A browser piano for toddlers.
Press a key, see it light up, hear it sing.

Musical Toy for Toddlers • Browser audio instrument • Made for toddlers and the grown-ups who play with them

Purpose

Help toddlers discover that their own actions can make music, right away.

PopNotes grows a child's motivation to explore and their sense of self-efficacy. There is nothing to read and no wrong answer. Every press leads to a sound, a light and a small success.

Characterisation

Approachable
Feels like a familiar toy, not software

Playful
Every press brings a little surprise

Encouraging
Every try counts as a success

Guiding
A friend shows the way, then steps aside

Design Rules

1. **One action, one instant response**
   Sound and light start the moment a key goes down.
2. **Colour before words**
   Each note keeps one colour everywhere.
3. **No wrong notes**
   A different key just means "try another one".
4. **Show, don't tell**
   A guide points and waits for the child to act.
5. **Keys never move**
   The toy scales as one piece on every screen.

Colour Palette

Bright key colours do the talking. Quiet neutrals let them stand out.

Key colours

#C43F43	#FC8B17	#FFE253	#69AA77	#17B3A6	#679ACA	#BF9CE6	#F899C0

Interface colours

Background	#EDE5DC	Warm cream page
Toy body	#DCDCDC	Holds every screen
Display	#F3F3F3	Light bars & notes
Outline	#898987	Key borders
Ink	#080808	Icons & text
Accent	#FAA64E	Selected & on
Marker	#FFFFFF	Next key dot

Typography

Few words, big and friendly.

Comic Sans MS

Display & interface · Bold

Aa

ABCDEFGHIJKLMNOPQRSTUVWXYZ
abcdefghijklmnopqrstuvwxyz 0123456789

Step titles, speech bubbles, song names and celebrations. Fallbacks: Chalkboard SE, Comic Neue. (Shown here in Comic Neue.)

Sans-serif

Key labels · Bold

A S D

Do Re Mi Fa Sol La Ti Do

Keyboard letters and solfege names on each key, plus small hints. Plain shapes stay clear at small sizes.

PopNotes wordmark

Logo · Custom lettering

PopNotes

A drawn title image, so it looks the same on every device. Chunky, rounded letters echo the toy shapes.

Type scale (all sizes scale with the toy, using cqw units)

So Good!	1. Try a key	Welcome	Twinkle Twinkle	Do Re Mi
Celebration	Step title	Speech bubble	Song name	Key label

Character

Meet the Guide: a friendly face that shows the way, then steps aside.

The Guide

A cheerful girl appears the first time a child visits each screen, shows one step at a time, and waits for the child to try it before moving on.

Headphones = music · Orange dress matches the accent · Pointing poses lead the eye · Big smile, no words needed

Poses

Introduce
"Welcome! Choose one to try"

Crouch
Show keyboard & note names

Point to key
Watch the note, press the key

Celebrate
"Try Playing!" finish card

Buttons & Icons

Every button gives a soft click sound.

Start
Enter the toy from the welcome screen

Free Play
Open play: press anything

Learning
Follow a song, note by note

Back / Home
Always top left, one step back

Volume
Mute everything with one press

Change Sound
Switch piano and xylophone

Shuffle Colours
Mix key colours, pitches stay

Label Switches
Show letters or Do Re Mi

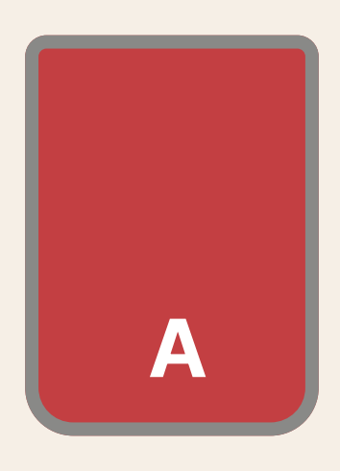
Song Library
Go back and pick another song

Song Cards
Twinkle · If You're Happy · Ode to Joy

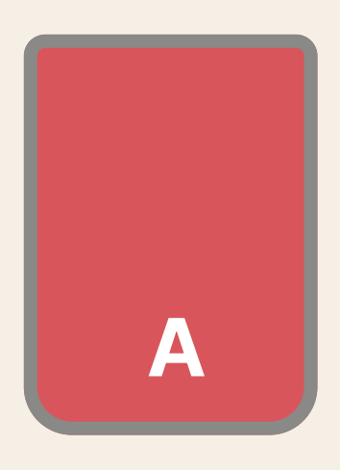
Listen
Hear the whole melody first

Celebrate
Finish cards and completed steps

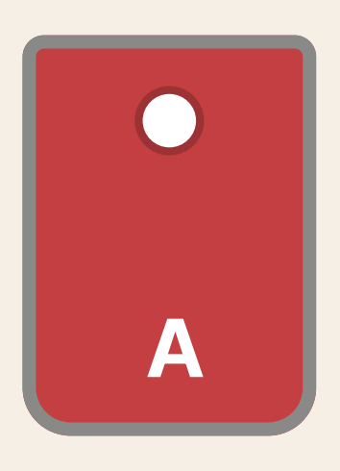
Key states



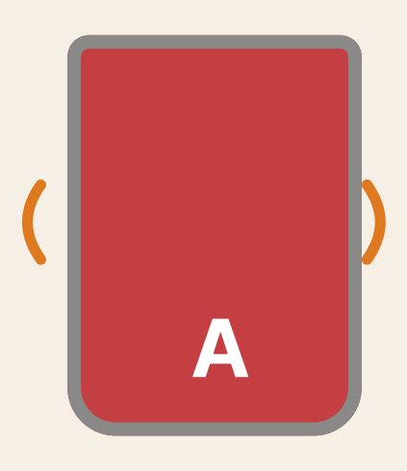
Default
Ready to play



Pressed
Sinks a little and brightens



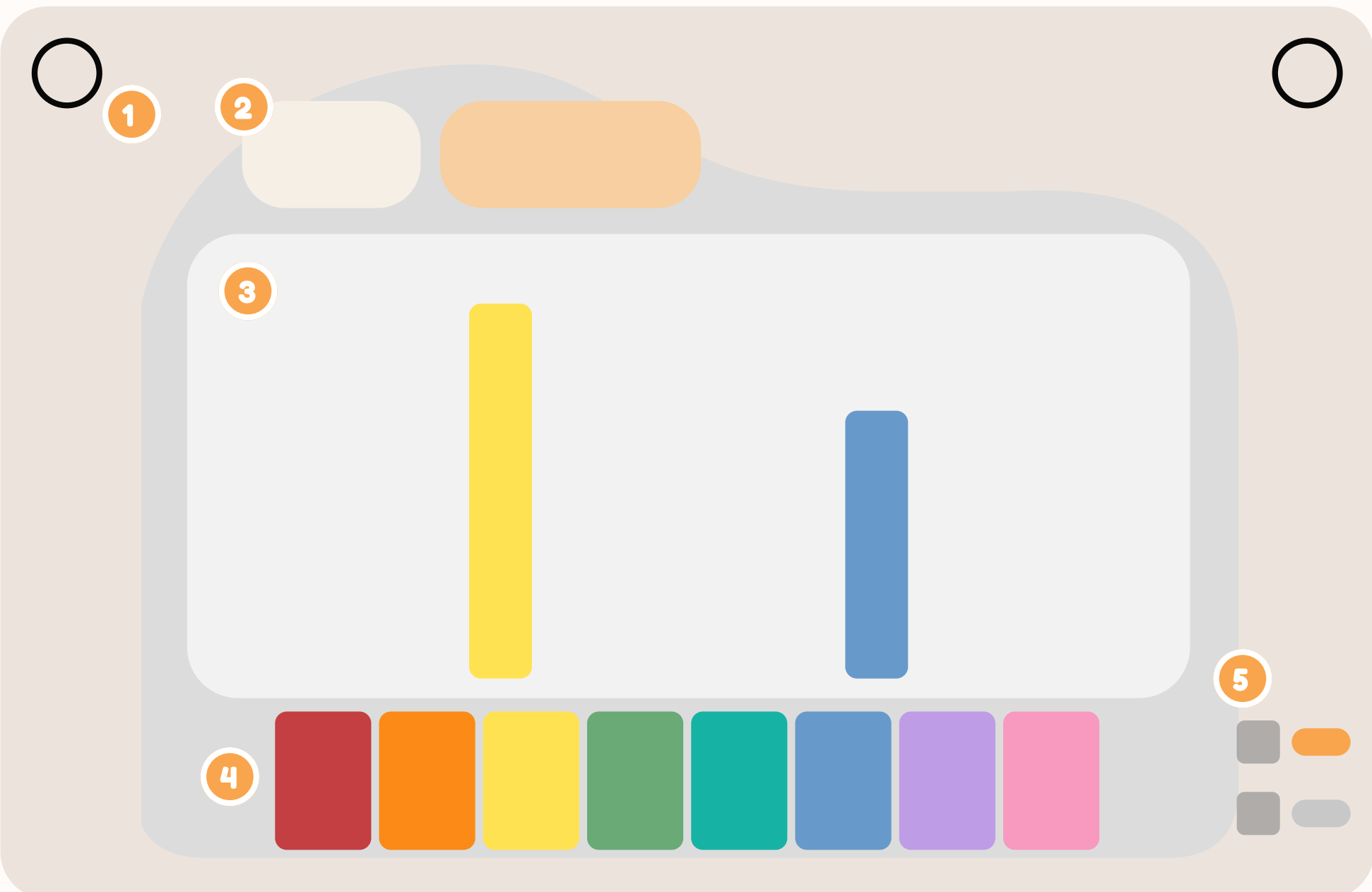
Next key
White dot shows where to go



Try again
Note still plays, a tiny shake

Layout

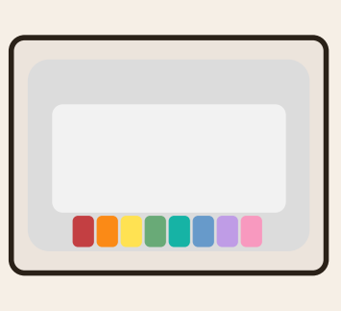
One toy, one view. Every screen sits inside the same rounded toy body, like a real toy piano.



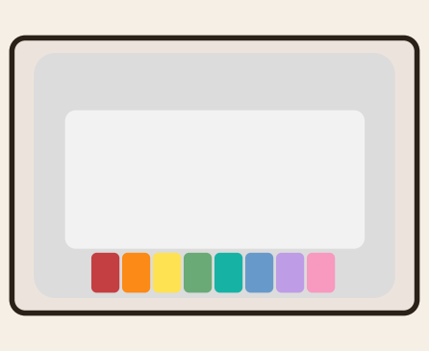
1. **Navigation**
   Back/Home top left, Volume top right. The only buttons that are always there.
2. **Controls**
   Tabs on top of the toy: Change Sound and Shuffle Colours, or the Song Library.
3. **Display**
   Light bars (Free Play) or falling notes (Learning). Eight lanes, each lined up with its key.
4. **Keys**
   Eight colourful keys along the bottom, in the same place on every screen.
5. **Label switches**
   Keyboard letters and Do Re Mi, tucked in the corner for the grown-up.

Responsive: the toy scales as one piece

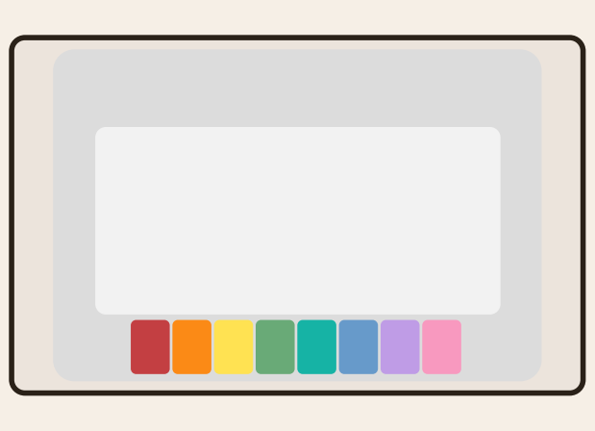
The toy keeps a 1340 : 870 shape. It fills the window's width but never grows taller than the window, and everything inside is sized with container units (cqw), so keys, text and icons scale together. Nav buttons stay between 44px and 82px. Tested in Chrome and Firefox.



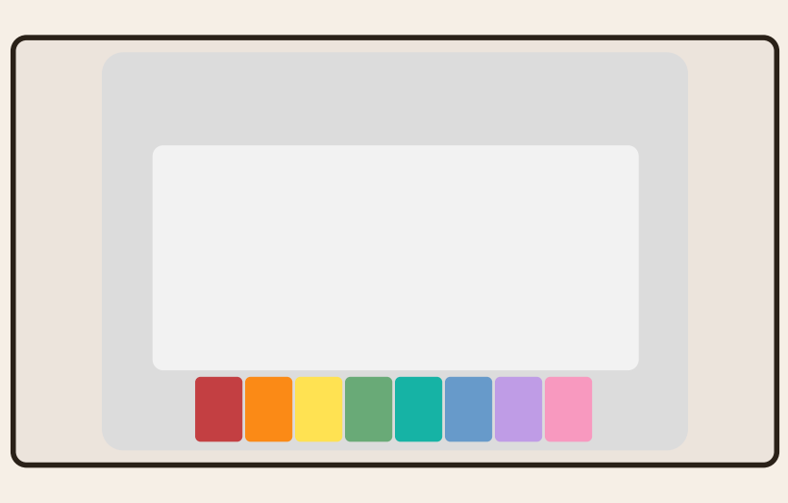
Medium
< 840px · 800×600
Toy fills the full width



Expanded
840 – 1199px · 1024×700
Scales up evenly



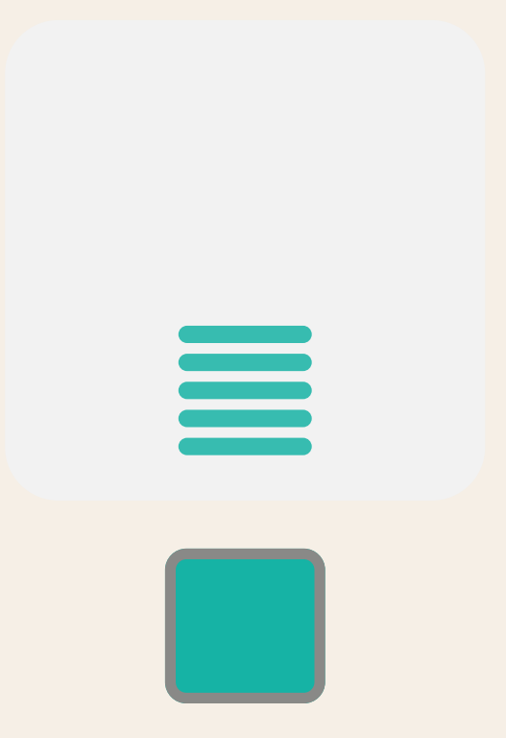
Large
1200 – 1600px · 1440×900
Height-limited, centred



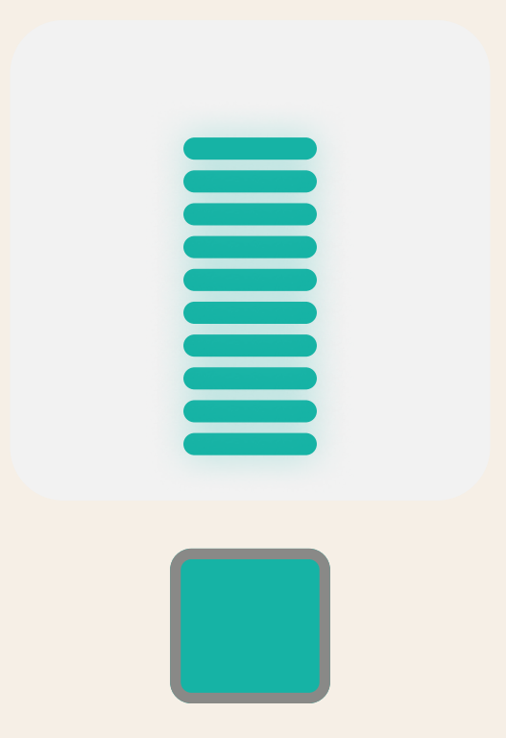
Extra-large
> 1600px · 1920×1080
Centred, cream fills the sides

Feedback & Motion

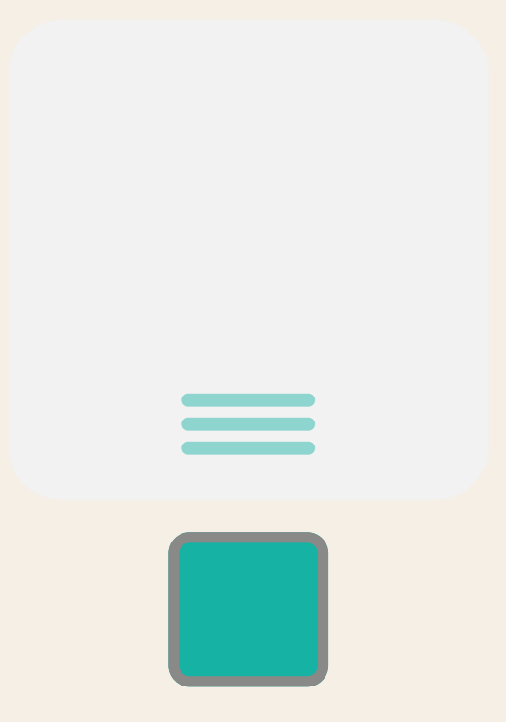
Movement is the clearest way to say "that happened because of you".



Press
The bar lights up at once



Hold
It rises and glows brighter



Release
It softly settles back

Timing

Key press → instant response · Bar rise 260–520 ms (random) · Glow pulse 1.1–1.8 s (random) · Try-again shake 0.22 s · Guide hand tap 1.2 s loop · Background float 24 s

PopNotes Design System · Values taken from the final build (css/style.css, js/audio.js, js/freeplay.js, js/learning.js)